

Lab3. Programming Paradigms Comparison

Student Name: Lastname Name

Group: _____

Date: _____

Objective

You will solve the same problem using different programming paradigms.

The goal is to understand how the way you think and structure a solution changes depending on the paradigm.

Required Paradigms

You must implement the solution using ALL of the following paradigms:

1. Imperative Programming
2. Structured Programming (functions + control structures)
3. Modular Programming
4. Object-Oriented Programming (OOP)
5. Functional Programming (basic level)

Important Rules

- The SAME problem must be used in all paradigms.
- You must NOT copy-paste the same solution with small changes.
- Each paradigm must show a different way of structuring the solution.
- Each section must include:
 - explanation in your own words
 - working code
 - a personal modification
 - a short reflection

Problem Description

Describe the problem you will solve:

A system that calculates the average grade of students and determines if they pass or fail.

✓ 1. 1. Imperative Programming

Brief Explanation

It's a paradigm to work step by step instructions using variables and loops.

Code Example

Complete the steps:

```
# Given data
grades = [70, 80, 90, 55, 95]
total = 0
highest = grades[0]
lowest = grades[0]
# 1. Calculate sum using a loop
for i in grades:
    total += i
    if i > highest:
        highest = i
    if i < lowest:
```

```

    lowest = i

average = total / len(grades)

print("promedio: ", average)
print("mas alta: ", highest)
print("mas baja: ", lowest)

if average >= 70:
    print("resultado aprobado")
else:
    print("resultado reprobado")

```

```

pass
Average: 80.0

```

Your Modification

What did you change and why? i added variables to find the highest and lowest grades within the same loop, without needing functions this way the program does more work but remains purely imperative one instruction after another Example: I added a function to calculate the highest grade and printed it together with the average.

Reflection

What did you learn from this approach and how does it compare to others? the imperative paradigm is easy to read when the program is small, but if the grades were for 30 different students, the code would be very repetitive, everything is mixed together the calculation, the logic, and the output are all in the same block compared to other paradigms, this one doesn't organize the code it just executes it it's useful for quick scripts but difficult to maintain

2. Structured Programming (functions + control structures)

✓ Brief Explanation

It uses functions to organize logic

Code Example

Complete the steps:

```

def calculate_mean(values):
    return sum(values) / len(values)

def checking_status(avg):
    if avg >= 70:
        return "Pass"
    return "Fail"

grades = [70, 55, 12]
avg = calculate_mean(grades)
status = checking_status(avg)
print("Average: ", avg)
print("Status: ", status)

```

```

Average: 45.666666666666664
Status: Fail

```

Your Modification

What did you change and why? i added a third function called get_highest() that finds the best grade in the list, and i also formatted the average to show only 2 decimal places using round

Reflection

What did you learn from this approach and how does it compare to others?

3.